

Total No. of Questions : 4]

SEAT No. :

P5100

[Total No. of Pages : 2

[6187]-507

T.E. (Electronics and Computer Engineering) (Insem)
DATABASE MANAGEMENT SYSTEM
(2019 Pattern) (Semester - I) (310341)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2 and Q.3 or Q.4.
- 2) Figures to the right indicate full marks.
- 3) Assume suitable data, if necessary.

Q1) a) What is entity? Explain strong entity and weak entity with example.
Compare strong entity and weak entity. [8]

b) What is relational algebra? Explain following operations of relational algebra with example. [7]

- i) Select operation
- ii) Union operation
- iii) Project Operation

OR

Q2) a) Explain Primary key and Foreign key with example. Write difference between Primary Key and Foreign Key. [8]

b) What is ER Model? Discuss with examples about various types of attributes present in the ER model. [7]

Q3) a) What does relational data model mean? Explain properties of relational database model. [5]

P.T.O.

Flexible
redundancy
security
represents
function

- b) Explain different features of good relational database design. [5]
- c) What is database anomaly? Explain insert and update anomaly with suitable example. [5]

OR

- Q4) a) Define functional dependency. Describe Trivial functional dependency and Non Trivial functional dependency with suitable example. [5]
- b) What is Normalization? State and explain 1NF & 2NF with example. [5]
- c) Explain any five Edgar Codd's Rules for relational database model. [5]

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Total No. of Questions : 6]

SEAT No. :

P8856

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Oct-22/TE/Insem-607

T.E. (Electronics and Computer Engineering)

DATABASE MANAGEMENT SYSTEM

(2019 Pattern) (Semester - I) (310341)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4 and Q5 or Q6.
- 2) Figures to the right side indicate full marks.
- 3) Assume suitable data, if necessary.

Q1) a) Draw ER diagram for library management system considering issue and return, fine collection facility. Consider appropriate entities. [6]

b) Explain how ER diagrams are converted into tables. [4]

OR

Q2) a) What is relational algebra? List and explain fundamental and additional relational algebra operators. [6]

b) Write advantages and disadvantages of Relational database model. [4]

Q3) a) Explain Cardinality in DBMS? Explain different types of relationship in DBMS. [6]

b) Define entity and entity set. What is E-R Model? [4]

OR

Q4) a) List all Codd's Rules and explain any two Rules in detail. [6]

b) Explain Data Redundancy and Update Anomalies with Example. [4]

P.T.O.

- Q5)** a) Explain basic Structure of Relational Model. [6]
b) Write a short note Multi valued Normalization with example. [4]

OR

- Q6)** a) Explain Decomposition along with its types and Example of Each. [6]
b) Explain the concept of Referential Integrity constraint and Entity Integrity Constraint with example. [4]

